

4.1 Warm-Up Exercises

In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and for the 3-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

4.1. Determine a Cartesian equations for the line ℓ in the following cases:

- a) ℓ contains the point $A(-2, 3)$ and has an angle of 60° with the Ox -axis,
- b) ℓ contains the point $B(1, 7)$ and is orthogonal to $\mathbf{n}(4, 3)$.

4.2. Consider a line ℓ . Show that

- a) if $\mathbf{v}(v_1, v_2)$ is a direction vector for ℓ then $\mathbf{n}(v_2, -v_1)$ is a normal vector for ℓ ,
- b) if $\mathbf{n}(n_1, n_2)$ is a normal vector for ℓ then $\mathbf{v}(n_2, -n_1)$ is a direction vector for ℓ .

How does this relate to the $\mathbf{J}$ -operator?

4.3. Let $A(1, 3)$, $B(-4, 3)$ and $C(2, 9)$ be the vertices of a triangle. Determine

- a) the length of the altitude from A ,
- b) the line containing the altitude from A .

4.4. Determine the angle between the lines $\ell_1 : y = 2x + 1$ and $\ell_2 : y = -x + 2$.

4.5. Determine Cartesian equations for the lines situated at distance 4 from the line $12x - 5y - 15 = 0$.

4.6. Determine the values k for which the distance from the point $(2, 3)$ to the line $8x + 15y + k = 0$ equals 5.

4.7. Consider the points $A(3, -1)$, $B(9, 1)$ and $C(-5, 5)$. For each pair of these three points, determine the line which is equidistant from them.

4.8. Determine a point on the line $5x - 4y - 4 = 0$ for which lies at equal distance from the points $A(1, 0)$ and $B(-2, 1)$.

4.9. Consider the vectors $\mathbf{a}(2, 1, 0)$ and $\mathbf{b}(0, -2, 1)$. Determine the angles between the diagonals of a parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

4.10. Consider the vectors $\mathbf{q} = 3\mathbf{i} + \mathbf{j}$ and $\mathbf{p} = \mathbf{i} + 2\mathbf{j} + \lambda\mathbf{k}$ with $\lambda \in \mathbb{R}$. Determine λ such that the cosine of the angle $\angle(\mathbf{p}, \mathbf{q})$ is $5/12$.

4.11. In an orthonormal basis, consider the vectors $\mathbf{v}_1(0, 1, 0)$, $\mathbf{v}_2(2, 1, 0)$ and $\mathbf{v}_3(-1, 0, 1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

4.12. Show that a parallelepiped with faces in the planes $2x + y - 2z + 6 = 0$, $2x - 2y + z - 8 = 0$ and $x + 2y + 2z + 1 = 0$ is rectangular.

4.13. Determine the distance between the planes $x - 2y - 2z + 7 = 0$ and $2x - 4y - 4z + 17 = 0$.

4.14. Determine the angles between the plane $x - \sqrt{2}y + z - 1 = 0$ and the plane $x + \sqrt{2}y - z + 3 = 0$.

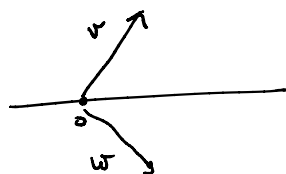
4.15. Determine the coordinates of a point A on the line $\ell : \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{1}$ which lies at distance $\sqrt{3}$ from the plane $x + y + z + 3 = 0$.

4.16. The vertices of a tetrahedron are $A(-1, -3, 1)$, $B(5, 3, 8)$, $C(-1, -3, 5)$ and $D(2, 1, -4)$. Determine the height of the tetrahedron relative to the face ABC .

4.1. Determine a Cartesian equations for the line ℓ in the following cases:

- ℓ contains the point $A(-2, 3)$ and has an angle of 60° with the Ox -axis,
- ℓ contains the point $B(1, 7)$ and is orthogonal to $\mathbf{n}(4, 3)$.

a.) there are two unit vectors forming a 60° angle with i
 $\mathbf{v}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ $\mathbf{w}\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$



$$\Rightarrow \ell: \frac{x+2}{1} = \frac{y-3}{\sqrt{3}} \quad \text{or} \quad \ell: \frac{x+2}{1} = \frac{y-3}{-\sqrt{3}}$$

or, if you prefer, use $\ell: y - y_A = \tan \alpha (x - x_A)$

$$y - 3 = \pm \sqrt{3} (x + 2)$$

b.) $\ell: n_x(x - x_0) + n_y(y - y_0) = 0$

so $\ell: 4(x - 1) + 3(y - 7) = 0 \Leftrightarrow \ell: 4x + 3y - 25 = 0$

4.2. Consider a line ℓ . Show that

- if $\mathbf{v}(v_1, v_2)$ is a direction vector for ℓ then $\mathbf{n}(v_2, -v_1)$ is a normal vector for ℓ ,
- if $\mathbf{n}(n_1, n_2)$ is a normal vector for ℓ then $\mathbf{v}(n_2, -n_1)$ is a direction vector for ℓ .

How does this relate to the J -operator?

a.) in dimension 2
 any vector orthogonal to a direction vector
 is a normal vector

$$\langle \mathbf{v}, \mathbf{n} \rangle = v_1 \cdot v_2 - v_2 \cdot v_1 = 0 \Rightarrow \mathbf{n} \perp \mathbf{v}$$

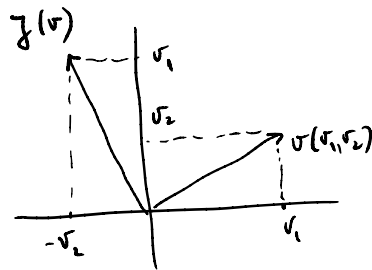
$\Rightarrow \mathbf{n}$ is a normal vector

here $\mathbf{n} = -J(\mathbf{v})$

b.) in dimension 2
 any vector orthogonal to a normal vector
 is a direction vector

$$\langle \mathbf{n}, \mathbf{v} \rangle = n_1 v_2 - n_2 v_1 = 0 \Rightarrow \mathbf{v} \perp \mathbf{n}$$

$\Rightarrow \mathbf{v}$ is a direction vector



here $\mathbf{v} = -J(\mathbf{n})$

4.3. Let $A(1, 3)$, $B(-4, 3)$ and $C(2, 9)$ be the vertices of a triangle. Determine

- the length of the altitude from A ,
- the line containing the altitude from A .

$$a) |AH| = d(A, BC) = \frac{|1-3+7|}{\sqrt{1+1}} = \frac{5}{\sqrt{2}}$$

$$\downarrow$$

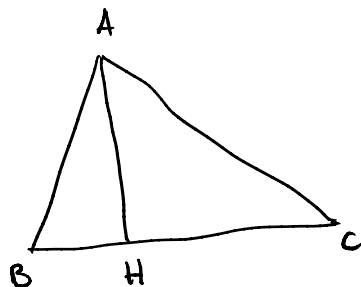
$$\vec{BC}(6, 6)$$



$$BC: \frac{x+4}{1} = \frac{y-3}{1} \Leftrightarrow x-y+7=0$$

$$b) (1, 1) \parallel \vec{BC}, \vec{BC} \perp \vec{AH} \Rightarrow (1, 1) \text{ is a normal vector for } AH$$

$$\Rightarrow AH: 1 \cdot (x-1) + 1 \cdot (y-3) = 0$$



4.4. Determine the angle between the lines $\ell_1: y = 2x + 1$ and $\ell_2: y = -x + 2$.

Sol 1

$$\tan \alpha = 2 \quad \tan \beta = -1 < 0$$

$\Rightarrow$ one angle is $\alpha + \beta$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{2-1}{1+2} = \frac{1}{3} \Rightarrow \alpha + \beta \approx 18.43^\circ$$

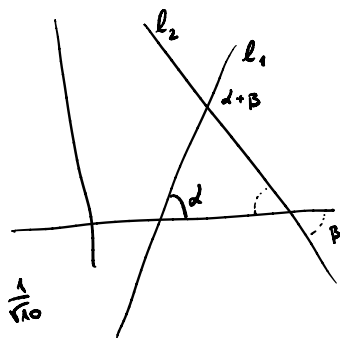
Sol 2

$n_1(2, -1)$ normal vector for ℓ_1

$n_2(1, 1)$ normal vector for ℓ_2

$$\Rightarrow \cos \angle(n_1, n_2) = \frac{\langle n_1, n_2 \rangle}{\|n_1\| \|n_2\|} = \frac{1}{\sqrt{10}}$$

$$\Rightarrow \angle(n_1, n_2) \approx 71.57^\circ$$



4.5. Determine Cartesian equations for the lines situated at distance 4 from the line $12x - 5y - 15 = 0$.

$$\text{For a point } P(x_0, y_0) \text{ we have } 4 = d(P, \ell) = \frac{|12x_0 - 5y_0 - 15|}{\sqrt{144 + 25}} = 13$$

$\Rightarrow$ the two lines are

$$\ell_{1,2}: 12x - 5y = 15 \pm 52$$

4.6. Determine the values k for which the distance from the point $(2,3)$ to the line $8x + 15y + k = 0$ equals 5.

$$5 = d(P, \ell) = \frac{|16 + 45 + k|}{\sqrt{64 + 225}} = 17 \quad \Leftrightarrow \quad 85 = \pm(61 + k)$$

$$\Rightarrow k = -146 \text{ or } k = 24$$

4.7. Consider the points $A(3, -1)$, $B(9, 1)$ and $C(-5, 5)$. For each pair of these three points, determine the line which is equidistant from them.

• These lines are the perpendicular bisectors of the sides of the triangle

Sol 1 For a point $P(x_0, y_0)$ we have

$$d(A, P) = d(B, P)$$

$$\Leftrightarrow d(A, P)^2 = d(B, P)^2$$

$$\Leftrightarrow (3 - x_0)^2 + (-1 - y_0)^2 = (9 - x_0)^2 + (1 - y_0)^2$$

$$\Leftrightarrow 9 - 6x_0 + x_0^2 + 1 + 2y_0 + y_0^2 = 81 - 18x_0 + x_0^2 + 1 - 2y_0 + y_0^2$$

$$\Leftrightarrow 12x_0 + 4y_0 - 72 = 0$$

$$\text{So } \ell: 3x_0 + y_0 - 18 = 0$$

Sol 2 $\vec{AB}(6, 2)$ is a normal vector for ℓ

$M(6, 0)$ is the midpoint of $[AB] \Rightarrow M \in \ell$

$$\Rightarrow \ell: 6(x - 6) + 2y = 0 \quad \Leftrightarrow \quad \ell: 3x + y - 18 = 0$$

4.8. Determine a point on the line $5x - 4y - 4 = 0$ which lies at equal distance from the points $A(1, 0)$ and $B(-2, 1)$.

Sol 1 $u(5, -4)$ normal vect for $\ell \Rightarrow v(4, 5) \parallel \ell \Rightarrow \ell: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4t \\ 5t + 1 \end{bmatrix} = P_t$

$$P(0, -1) \in \ell$$

$$d(P_t, A) = d(P_t, B) \Leftrightarrow \dots \text{write out eq. and solve in } t \dots t = -\frac{3}{7}$$

Sol 2 The point P which we seek is $\ell \cap \ell'$ where ℓ' is the perp. bisector of segment $[AB]$

$$\vec{AB}(-3, 1), \quad (-\frac{1}{2}, \frac{1}{2}) \text{ mid point of } [AB]$$

$$\Rightarrow \ell': -3\left(x + \frac{1}{2}\right) + 1 \cdot \left(y - \frac{1}{2}\right) = 0$$

$$\Rightarrow \ell': -3x + y - 2 = 0$$

$$\ell \cap \ell': \begin{cases} 5x - 4y - 4 = 0 \\ 3x - y + 2 = 0 \end{cases} \Rightarrow y = 3x + 2 \Rightarrow \begin{cases} 5x - 12x - 8 - 4 = 0 \\ -7x - 12 = 0 \end{cases} \Rightarrow x = -\frac{12}{7}$$

$$\Rightarrow y = -\frac{12}{7}$$

4.9. Consider the vectors $\mathbf{a}(2, 1, 0)$ and $\mathbf{b}(0, -2, 1)$. Determine the angles between the diagonals of a parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

$$\mathbf{d}_1 = \mathbf{a} + \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{d}_2 = \mathbf{a} - \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \cos \angle(\mathbf{d}_1, \mathbf{d}_2) = \frac{\langle \mathbf{d}_1, \mathbf{d}_2 \rangle}{|\mathbf{d}_1| \cdot |\mathbf{d}_2|} = \frac{4 - 3 - 1}{\dots} = 0$$

$$\Rightarrow \angle(\mathbf{d}_1, \mathbf{d}_2) = 90^\circ$$

4.10. Consider the vectors $\mathbf{q} = 3\mathbf{i} + \mathbf{j}$ and $\mathbf{p} = \mathbf{i} + 2\mathbf{j} + \lambda\mathbf{k}$ with $\lambda \in \mathbb{R}$. Determine λ such that the cosine of the angle $\angle(\mathbf{p}, \mathbf{q})$ is $5/12$.

$$\mathbf{q} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{p} = \begin{bmatrix} 1 \\ 2 \\ \lambda \end{bmatrix} \quad \cos \angle(\mathbf{p}, \mathbf{q}) = \frac{3 + 2 + \lambda \cdot 0}{\sqrt{10} \cdot \sqrt{5 + \lambda^2}} = \frac{5}{12}$$

$$\Leftrightarrow \sqrt{10} \cdot \sqrt{5 + \lambda^2} = 12 \Leftrightarrow 10(5 + \lambda^2) = 144 \Leftrightarrow \lambda^2 = \frac{34}{10}$$

$$\Rightarrow \lambda = \pm \sqrt{3.4}$$

4.11. In an orthonormal basis, consider the vectors $\mathbf{v}_1(0, 1, 0)$, $\mathbf{v}_2(2, 1, 0)$ and $\mathbf{v}_3(-1, 0, 1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

$$\mathbf{w}_1 = \mathbf{v}_1$$

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{w}_1, \mathbf{v}_2 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \frac{\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \rangle}{1^2} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{w}_3 = \mathbf{v}_3 - \frac{\langle \mathbf{w}_1, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 - \frac{\langle \mathbf{w}_2, \mathbf{v}_3 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rangle}{1^2} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{\langle \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rangle}{4} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - 0 + \frac{1}{2} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\Rightarrow \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ is an orthogonal basis

$\Rightarrow$ an orthonormal basis is $\frac{\mathbf{w}_1}{|\mathbf{w}_1|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \frac{\mathbf{w}_2}{|\mathbf{w}_2|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \frac{\mathbf{w}_3}{|\mathbf{w}_3|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

4.12. Show that a parallelepiped with faces in the planes $2x + y - 2z + 6 = 0$, $2x - 2y + z - 8 = 0$ and $x + 2y + 2z + 1 = 0$ is rectangular.

- A parallelepiped is rectangular if its faces are orthogonal to each other
 $(\Rightarrow)$ the planes containing the faces have 90° dihedral angles
- normal vectors of the given planes are $n_1(2, 1, -2)$
 $n_2(2, -2, 1)$
 $n_3(1, 2, 2)$

and one checks that $\langle n_1, n_2 \rangle = 0$, $\langle n_1, n_3 \rangle = 0$, $\langle n_2, n_3 \rangle = 0$

4.13. Determine the distance between the planes $\underbrace{x - 2y - 2z + 7 = 0}_{\pi_1}$ and $\underbrace{2x - 4y - 4z + 17 = 0}_{\pi_2}$.

a normal vector for π_1 is $n_1(1, -2, -2)$
 a normal vector for π_2 is $n_2(2, -4, -4)$ $\Rightarrow n_1 \parallel n_2 \Rightarrow \pi_1 \parallel \pi_2$

$\Rightarrow d(\pi_1, \pi_2) = d(\pi_1, P) \quad \forall P \in \pi_2$
 choose $P = (-\frac{17}{2}, 0, 0)$ for example. Then $d(\pi_1, \pi_2) = \frac{|-\frac{17}{2} - 2 \cdot 0 - 2 \cdot 0 + 7|}{\sqrt{1+4+4}} = \frac{\frac{3}{2}}{\sqrt{5}} = \frac{3}{2\sqrt{5}}$

4.14. Determine the angles between the plane $\underbrace{x - \sqrt{2}y + z - 1 = 0}_{\pi_1}$ and the plane $\underbrace{x + \sqrt{2}y - z + 3 = 0}_{\pi_2}$.

$n_1(1, -\sqrt{2}, 1)$ is a N.V for π_1

$n_2(1, \sqrt{2}, -1)$ —||— π_2

$\cos \angle(n_1, n_2) = \frac{\langle n_1, n_2 \rangle}{|n_1| \cdot |n_2|} = \frac{1 - 2 - 1}{\sqrt{4} \sqrt{4}} = -\frac{2}{4} = -\frac{1}{2} \Rightarrow \angle(n_1, n_2) = 120^\circ$
 $\Rightarrow$ the other angle is 60°

4.15. Determine the coordinates of a point A on the line $\ell: \frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{1}$ which lies at distance $\sqrt{3}$ from the plane $\underbrace{x + y + z + 3 = 0}_{\pi}$.

the points on ℓ are $P_t = (2t+1, 3t, t-1)$ for $t \in \mathbb{R}$

$\Rightarrow \sqrt{3} = d(P, \pi) = \frac{|2t+1 + 3t + t-1 + 3|}{\sqrt{3}} \Rightarrow 3 = |6t+3| \Rightarrow 1 = |2t+1|$ $\begin{cases} 1 = 2t+1 \Rightarrow t=0 \\ 1 = -2t-1 \Rightarrow t=-1 \end{cases}$

$\Rightarrow$ the two points are $(1, 0, -1)$ and $(-1, -3, -2)$

4.16. The vertices of a tetrahedron are $A(-1, -3, 1)$, $B(5, 3, 8)$, $C(-1, -3, 5)$ and $D(2, 1, -4)$. Determine the height of the tetrahedron relative to the face ABC .

$$\overline{d(D, ABC)} \quad \overrightarrow{AB} = \begin{pmatrix} 6 \\ 6 \\ 7 \end{pmatrix} \quad \overrightarrow{AC} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} \Rightarrow ABC: \begin{vmatrix} x+1 & y+3 & z-1 \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{vmatrix} = 0$$

$$24(x+1) - 24(y+3) = 0$$

$$\Rightarrow ABC: x - y - 2 = 0$$

$$\Rightarrow d(D, ABC) = \frac{|2 - 1 - 2|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$